# Recommendations for Life History Data Needs for US Caribbean Fisheries Stock Assessment

## Summary

A primary goal of the Southeast Fisheries Science Center's Caribbean Strategic Planning project is to reduce gaps in life history datasets used to inform management. The first step in accomplishing that goal is to identify major life history data gaps and information needed to improve fisheries management decisions in the U.S. Caribbean. A working group was established to identify priority species for the U.S. Caribbean, identify available life history data including historical and ongoing efforts, summarize the identified life history data sources, identify life history data needs by species and spatial extent, and provide recommendations to the stakeholders to implement data collection that fill the gaps and inform stock assessment using the most currently available data. This document contains those findings.

A deliverable of the Caribbean Strategic Planning Project
Updated June 2026

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## Acronyms

| *Acronym* | *Definition* |
|---|---|
| CIMAS | Cooperative Institute for Marine and Atmospheric Studies |
| CRP | Cooperative Research Program grant |
| FMP | Fishery Management Plan |
| MARFIN | Marine Fisheries Initiative grant |
| PR | Puerto Rico |
| PR DRNA | Puerto Rico Departamento de Recursos Naturales y Ambientales |
| SEAMAP-C | Southeast Area Monitoring and Assessment Program - Caribbean |
| SEDAR | Southeast Data Assessment and Review |
| SEFSC | Southeast Fisheries Science Center |
| SK | Saltonstall-Kennedy grant |
| STJ | St. John |
| STT | St. Thomas |
| USC-A | University of South Carolina-Aiken |
| USVI | United States Virgin Islands |

## Working Group

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## Introduction

The data essential to complete a fisheries stock assessment include the abundance, biology, and catch for the species. Biological data refers to life history demographics including parameters for weight-length relationships, age-length relationships, maximum age, growth, maturity, and mortality. These parameters are inputted into the selected assessment model to inform the health of the population given the level of catch observed in the fishery. It is preferred that the life history parameters are obtained through research conducted on samples from the region in which the stock is being assessed, rather than from other regions (i.e., for a Puerto Rico stock assessment, it is best if life history parameters come from Puerto Rico rather than Florida). If data from the region are not available at the time of the assessment, estimates are made using data from nearby regions. This increases the uncertainty in the assessment model and the results may not be as informative for management. This working group was formed to strengthen the stock assessment enterprise in the U.S. Caribbean and improve the data available for stock assessment by identifying species-specific life history data gaps in the region.

Participants volunteered to join this strategic planning working group. During the initial meeting of the group, the participants developed their purpose, goals, and ideal outcomes, which are documented in the Team Charter. The group met sixteen times over the course of two years to accomplish the goals. Section A contains those goals and what was accomplished under each goal. Section B builds on the accomplished goals and contains specific recommendations for life history data collection.

## A. Goals & Summary of Actions to Address Goals:

1. Identify priority species for the U.S. Caribbean

   Many species are included in the island-based FMPs, but in tackling this project it was important to identify which species were considered priority in terms of identifying available life history data. The SEFSC Caribbean Fisheries Branch Data Triage Project categorized species as having a likelihood for a successful stock assessment based on the availability of the data required for the assessment (e.g., abundance, biology, and catch). Results of the data triage project were referenced to identify priority species lists for each island platform (i.e. Puerto Rico, St. Thomas/St. John, and St. Croix). The island-based FMPs were also referenced and species assigned as “indicator” were automatically included on the lists. Species in which ongoing life history research is underway were also included on the priority lists.

2. Identify available life history data for priority species

   Group discussion and literature review indicated that existing inventories of Caribbean life history data sources were non-comprehensive. The SEFSC Caribbean Fisheries Branch Data Triage Project and Stevens et al., 2019 have addressed cataloging reliable life history data sources and parameters in the past. Other forums have also worked to identify life history data sources including the SEFSC Caribbean Research Inventory and Fish/Fisheries Conservation Lab at USC-A. These efforts have identified a number of reliable life history sources available at the time, but a comprehensive catalog was still needed.

A new and comprehensive inventory was compiled to catalog reliable life history data for U.S. Caribbean priority species by referencing the previous efforts, performing additional literature searches, and soliciting feedback from life history researchers in the region. The inventory sought to capture the publication link, species, sampling location and years, depth range of sampling, gear used, sample size, and type of life history data collected and analyzed such as length age, length length, length weight, reproduction/maturity, and mortality. See Table 1 for the list of life history publications and the associated spreadsheet for additional information.

3. Summarize available data and identify life history data needs by species and spatial extent

The data gap identification process involved many steps. First, separate tables were created for each island platform and included each of the associated priority species. Next, it was noted which species had an assessment attempted in the SEDAR process. Throughout SEDAR, the lead analyst will compile a list of recommendations as to what data, if they were available, would improve the assessment modeling and better inform the management advice. These recommendations are listed at the end of the assessment report and were included in the tables. Next, the SEDAR recommendations and inventory of published data were used to summarize the current status of available life history data and the data collected and made available since the previous assessment. For species in which a SEDAR has not been attempted, the inventory was used to inform the status of available data. See Tables 2–4 for the information on SEDAR recommendations, overall status of available data, and availability of reproduction and age/growth data. Finally, a table highlighting the existing data gaps was produced. See Table 5.

4. Provide recommendations to the stakeholders to implement data collections that fill the data gaps and inform stock assessment using the most current available data

This working group concisely documented what data collections are needed to inform stock assessment and management advice. This present report, specifically section C, contains the findings of this group.

Table 1. Life history publications of priority species in the U.S. Caribbean.

| SPECIES | PLATFORM | YEARS | DATA TYPE | PUBLICATION |
|---|---|---|---|---|
| Queen triggerfish | PR, USVI | 2013-2023 | Length age, reproduction/maturity | Rivera-Hernández & Shervette, 2024 |
| Stoplight parrotfish, queen parrotfish | PR, USVI | 2018-2019 | Length age | Rivera-Hernández & Shervette, 2024 |
| Stoplight parrotfish, queen parrotfish | PR, USVI | 2015-2023 | Length age, reproduction/maturity | Rivera-Hernández & Shervette, 2024 |
| Yellowtail snapper | PR, USVI | 2009-2023 | Length age, length length, length weight, reproduction/maturity | Shervette et al., 2024 |
| Yellowtail snapper | PR, USVI | 2010-2023 | Length age, length weight | Shervette & Rivera Hernandez, 2024 |
| Red hind | USVI | 2018-2019 | Reproduction/maturity | Rosemond et al., 2022 |
| Queen triggerfish | PR, USVI | 2013-2021 | Length age | Shervette & Rivera Hernandez, 2022 |
| Caribbean spiny lobster | Wider Caribbean | 2017-2019 | Reproduction/maturity | Atherley et al., 2021 |
| Princess parrotfish | PR, USVI | 2015-2020 | Length age, reproduction/maturity | Jones et al., 2021 |
| Yellowfin grouper | STT | 2008, 2009, 2011, 2014 | Reproduction/maturity | Nemeth et al., 2020 |
| Queen triggerfish | PR, STX | 2013-2018 | Reproduction/maturity | Rivera-Hernández et al., 2018 |
| Caribbean spiny lobster | STT | 2012-2016 | Length age | Olsen et al., 2017 |
| Caribbean spiny lobster | Wider Caribbean | 2015-2016 | Reproduction/maturity | Tewfik & Babcock, 2017 |
| Blackfin snapper | PR, USVI | 1979-2015 | Length weight | Burton et al., 2016 |
| Hogfish | Wider Caribbean | 2011-2013 | Length length, length weight | Pérez-Chacón, 2015 |
| Yellowtail snapper | USVI | 1974-2012 | Length age, mortality | Olsen, 2014 |
| French grunt, schoolmaster snapper | PR, STX | 2007 | Length age | Mateo et al., 2011 |
| Stoplight parrotfish | Wider Caribbean | | Length length | Molloy et al. 2011 |
| Yellowfin grouper | STT | 2004-2009 | Reproduction/maturity | Kadison et al., 2010 |
| Wahoo | Wider Caribbean | 1997-2006 | Reproduction/maturity | Maki Jenkins & McBride, 2009 |
| Snappers, groupers, hogfish | PR | 1985-1987, 2000-2005 | Length age, reproduction/maturity, mortality | Ault et al., 2008 |
| King mackerel, cero mackerel | PR | 2001-2002 | Reproduction/maturity | Figuerola-Fernández et al., 2007 |
| Caribbean spiny lobster, queen conch | Wider Caribbean | | | Posada & Glazer, 2007 |

| | | | | |
|---|---|---|---|---|
| Silk snapper | PR | 2005 | Reproduction/maturity | Rosario et al., 2006 |
| Reef fish | STT | 2005-2006 | Length weight | Trumble et al., 2006 |
| Caribbean spiny lobster | PR | 1980-2003 | | Chormanski et al., 2005 |
| Queen snapper | Wider Caribbean | 1986-1987, 1992, 1999-2000, 2001 | Length length, length weight, Reproduction/maturity | Gobert et al., 2005 |
| Red hind | USVI | 1999-2004 | Reproduction/maturity | Nemeth, 2005 |
| Dolphinfish | PR | 1990-1992 | Reproduction/maturity | Perez & Roman, 2005 |
| Red hind | STT | 2002-2003 | Reproduction/maturity | Whiteman et al., 2005 |
| Stoplight parrotfish | Wider Caribbean | 1995-2000 | Length age | Choat et al. 2003 |
| Wahoo | Wider Caribbean | 1976-2000 | Length age, length weight, Reproduction/maturity | Oxenford et al., 2003 |
| Parrotfish, surgeonfish | PR, USVI | 1991-2000 | Length weight | Pagan, 2002 |
| Coney grouper | PR | 1997-1999 | Reproduction/maturity | Figuerola et al., 2001 |
| Mutton snapper | PR | 2000-2002 | Reproduction/maturity | Figuerola & Torres., 2001 |
| Wahoo | Wider Caribbean | 1997-1998 | Reproduction/maturity | Brown-Peterson et al., 2000 |
| Wahoo | Wider Caribbean | 1997-1998 | Length age | Franks et al., 2000 |
| Lane snapper | Wider Caribbean | 1992-1996 | Length age | Luckhurst et al., 2000 |
| Dolphinfish | PR | 1991-1992 | Length age, length weight | Rivera & Appeldoorn, 2000 |
| Caribbean spiny lobster | STX | 1995-1999 | Length age | Mateo & Tobias, 1999 |
| Groupers, parrotfish, snappers | PR | 1996-1998 | Reproduction/maturity | Figuerola et al., 1998 |
| Stoplight parrotfish | Wider Caribbean | 1989-1992 | Mortality | van Rooij & Videler, 1997 |
| Mutton snapper | Wider Caribbean | 1991-1992 | Length length | Mueller, 1995 |
| Coney grouper, red hind | USVI | 1978-1992 | Length weight | Bolden, 1994 |
| Dolphinfish | PR | | Length age, length weight | Rivera Betancourt, 1994 |
| Red hind | PR | 1987-1992 | Reproduction/maturity | Sadovy et al., 1994 |
| Red hind | PR | 1982-1989 | Length weight, reproduction/maturity | Shapiro et al., 1993 |
| Lane snapper | PR | 1988-1988 | Length age, length weight, mortality | Acosta & Appeldoorn., 1992 |
| Red hind | USVI | 1984-1991 | Length weight | Beets & Friedlander, 1992 |
| Red hind | Wider Caribbean | 1990 | Length weight | Luckhurst, 1992 |
| Dolphinfish | PR | 1990-1992 | Length weight, reproduction/maturity | Pérez et al., 1992 |

| | | | | |
|---|---|---|---|---|
| Red hind | PR, STT | 1987-1989 | Length weight, reproduction/maturity | Sadovy et al., 1992 |
| Silk snapper | PR | 1989-1991 | Reproduction/maturity | Figuerola, 1991 |
| Dolphinfish | PR | 1990-1991 | Length weight, reproduction/maturity | Perez & Sadovy, 1991 |
| White grunt | PR | 1988-1989 | | Román, 1991 |
| Red hind, white grunt | PR, STT | | Length age | Sadovy et al., 1989 |
| Reef fish | PR, USVI | 1985 | Length weight | Bohnsack & Harper, 1988 |
| Lane snapper | Wider Caribbean | 1979-1981 | Length age, reproduction/maturity | Manickchand-Dass, 1987 |
| Yellowtail snapper, queen triggerfish | PR, USVI | 1983-1984 | Length age, length weight | Manooch & Drennon, 1987 |
| Yellowfin grouper | STJ | | Length age | Munro & Williams, 1985 |
| Hogfish | PR | | Reproduction/maturity | Colin, 1982 |
| Snappers | PR | 1976-79 | Length weight, reproduction/maturity | Boardman & Weiler, 1980 |
| White grunt | PR | 1973 | Length age | Stevenson, 1978 |
| Silk snapper | USVI | 1970-1972 | Length weight | Sylvester & Dammann, 1973 |
| Groupers | STJ | 1959-1960 | Length age | Randall, 1962 |

Table 2. Comprehensive list of available life history data for priority species in Puerto Rico.

| | | | | *AVAILABLE DATA* | |
|---|---|---|---|---|---|
| *SPECIES* | *SEDAR* | *SEDAR DATA RECOMMENDATION* | *STATUS OF AVAILABLE DATA* | *REPRODUCTION* | *AGE / GROWTH* |
| Bar jack | – | – | Non-comprehensive | Scarce data from SEAMAP-C | – |
| Blackfin tuna | – | – | Non-comprehensive | – | – |
| Blue runner | – | – | Non-comprehensive | Final report from F48 (2003-07) and data from SEAMAP-C | – |
| Boxfishes | – | – | Non-comprehensive | Scarce data from SEAMAP-C (only L. trigonus) /Final Report Yvone Sadovy, 1990 | Scarce data from SEAMAP-C (only L. trigonus) /Final Report Yvone Sadovy, 1990 |
| Caribbean spiny lobster | SEDAR8<br><br>SEDAR57<br><br>SEDAR91 | Mortality of juveniles<br>Size composition<br><br>Stock structure and abundance<br>Size composition<br>Life history<br><br>[IN PROGRESS] | Non-comprehensive | – | – |
| Cero mackerel | – | – | Non-comprehensive | Figuerola-Fernández et al., 2007 | – |
| Coney grouper | – | – | Non-comprehensive | Some reproductive histology data from SEAMAP-C samples | – |
| Dolphinfish | – | – | Non-comprehensive | DRNA opportunistic tournament sampling/Perez & Sadovy, 1991 | Rivera & Appeldoorn, 2000 |
| Graysby grouper | – | – | Non-comprehensive | Some reproductive histology data from SEAMAP-C samples | – |
| Hogfish | SEDAR46 | Length composition<br>Life history | Comprehensive | CRP report 2015-2021 collections for life history | CRP report 2015-2021 collections for life history |
| King mackerel | – | – | Non-comprehensive | Figuerola-Fernández et al., 2007 | – |
| Lane snapper | – | – | Comprehensive | 2012-2023 collections for life history; Figuerola et al., 1998 | 2012-2023 collections for life history |
| Little tunny | – | – | Non-comprehensive | – | – |

| | | | | | |
|---|---|---|---|---|---|
| Mutton snapper | SEDAR14 | Size composition<br>Age composition<br>Life history | Non-comprehensive | Figuerola & Torres., 2001; Data from 2016-2022 | – |
| Queen conch | SEDAR14 | Size composition<br>Age composition<br>Habitat | Non-comprehensive | SK 2024-2026 collections for life history | SK 2024-2026 collections for life history |
| Queen parrotfish | – | – | Comprehensive | Life history data available from 2013-2023; Rivera-Hernández & Shervette, 2024 | Life history data available from 2013-2023; Rivera-Hernández & Shervette, 2024 |
| Queen snapper | SEDAR4<br><br>SEDAR26 | [NONE]<br><br>Age and growth<br>Life history<br>Size composition | Non-comprehensive | SK 2023-2025 life history study underway | SK 2023-2025 life history study underway |
| Queen triggerfish | SEDAR30<br><br><br><br>SEDAR80 | Size composition<br>Age composition<br>Life history<br><br>Growth<br>Life history | Comprehensive | Rivera-Hernandez & Shervette, 2024 covers 2013-2023 | Rivera-Hernandez & Shervette, 2024 covers 2013-2023 |
| Rainbow parrotfish | – | – | Non-comprehensive | – | – |
| Rainbow runner | – | – | Non-comprehensive | – | – |
| Red hind grouper | SEDAR35 | Size composition<br>Age composition<br>Life history | Non-comprehensive | Some reproductive histology data from SEAMAP-C samples | – |
| Redtail parrotfish | SEDAR26 | Size composition<br>Age composition<br>Age and growth<br>Life history | Comprehensive | Life history data available from 2013-2023; MARFIN 2021 report; currently processing 2022-2023 samples; Figuerola et al., 1998 | Life history data available from 2013-2023; MARFIN 2021 report; currently processing 2022-2023 samples |
| Silk snapper | SEDAR26 | Size composition<br>Age composition<br>Age and growth<br>Life history | Non-comprehensive | SK 2023-2025 life history study underway | SK 2023-2025 life history study underway |
| Skipjack tuna | – | – | Non-comprehensive | – | – |
| Stoplight parrotfish | SEDAR84 | [IN PROGRESS] | Comprehensive | Life history data available from 2013-2023; SEDAR 84 2013-2023; Rivera-Hernández & Shervette, 2024; Figuerola et al., 1998 | Life history data available from 2013-2023, SEDAR 84 2013-2023; Rivera-Hernández & Shervette, 2024 |

| | | | | | |
|---|---|---|---|---|---|
| Wahoo | – | – | Non-comprehensive | DNER opportunistic tournament sampling/Final Report F48 (2004-2009) | – |
| White grunt | – | – | Non-comprehensive | Some reproductive histology data from SEAMAP-C samples; Román, 1991 | Sadovy et al., 1989 |
| Yellowfin tuna | – | – | Non-comprehensive | – | – |
| Yellowtail snapper | SEDAR8<br><br><br><br><br><br><br>SEDAR46<br><br><br>SEDAR84 | Movement patterns<br>Growth<br>Life history<br>Mortality<br>Size composition<br>Age composition<br><br>Length composition<br>Life history<br><br>[IN PROGRESS] | Comprehensive | SEDAR 84 Report (working paper); Figuerola et al., 1998 | SEDAR 84 Report (working paper) |

Table 3. Comprehensive list of available life history data for priority species in St. Thomas / St. John.

| | | | | *AVAILABLE DATA* | |
|---|---|---|---|---|---|
| *SPECIES* | *SEDAR* | *SEDAR DATA RECOMMENDATION* | *STATUS OF AVAILABLE DATA* | *REPRODUCTION* | *AGE / GROWTH* |
| Blackfin snapper | – | – | Non-comprehensive | – | – |
| Blue runner | – | – | Non-comprehensive | – | – |
| Blue tang | SEDAR30 | Size composition<br>Age composition<br>Life history | Non-comprehensive | – | – |
| Boxfishes | – | – | Non-comprehensive | – | – |
| Caribbean spiny lobster | SEDAR8<br><br>SEDAR46<br><br>SEDAR57<br><br><br><br>SEDAR91 | Mortality of juveniles<br>Size composition<br><br>Length composition<br>Life history<br><br>Stock structure and abundance<br>Size composition<br>Life history<br><br>[IN PROGRESS] | Non-comprehensive | – | – |
| Cero mackerel | – | – | Non-comprehensive | – | – |
| Coney grouper | – | – | Non-comprehensive | – | – |
| Doctorfish | – | – | Non-comprehensive | – | – |
| Dolphinfish | – | – | Non-comprehensive | – | – |
| Gray angelfish | – | – | Non-comprehensive | – | – |
| Graysby grouper | – | – | Non-comprehensive | – | – |
| Hogfish | – | – | Non-comprehensive | – | – |
| King mackerel | – | – | Non-comprehensive | – | – |
| Lane snapper | – | – | Comprehensive | 2012-2023 collections for life history | 2012-2023 collections for life history |
| Little tunny | – | – | Non-comprehensive | – | – |
| Margate grunt | – | – | Non-comprehensive | – | – |

| | | | | | |
|---|---|---|---|---|---|
| Mutton snapper | SEDAR14 | Size composition<br>Age composition<br>Life history | Non-comprehensive | – | – |
| Queen angelfish | – | – | Non-comprehensive | – | – |
| Queen conch | SEDAR14 | Size composition<br>Age composition<br>Habitat | Non-comprehensive | – | – |
| Queen snapper | SEDAR26 | Age and growth<br>Life history<br>Size composition | Non-comprehensive | SK 2023-2025 life history study underway | SK 2023-2025 life history study underway |
| Queen triggerfish | SEDAR30<br><br><br>SEDAR46<br><br>SEDAR80 | Size composition<br>Age composition<br>Life history<br><br>Length composition<br>Life history<br><br>[IN PROGRESS] | Comprehensive | Rivera-Hernández & Shervette, 2024 covers 2013-2023 | Rivera-Hernández & Shervette, 2024 covers 2013-2023 |
| Rainbow runner | – | – | Non-comprehensive | – | – |
| Red hind grouper | SEDAR35 | Size composition<br>Age composition<br>Life history | Non-comprehensive | CRP 2024-2025 study initiated | CRP 2024-2025 study initiated |
| Redtail parrotfish | SEDAR26 | Size composition<br>Age composition<br>Age and growth<br>Life history | Comprehensive | Life history data available from 2013-2023; MARFIN 2021 report; currently processing 2022-2023 samples | Life history data available from 2013-2023; MARFIN 2021 report; currently processing 2022-2023 samples |
| Scrawled cowfish | – | – | Non-comprehensive | – | – |
| Stoplight parrotfish | – | – | Comprehensive | Life history data available from 2013-2023; SEDAR 84 2013-2023; Rivera-Hernández & Shervette, 2024; Rivera-Hernández & Shervette, 2024 | Life history data available from 2013-2023; SEDAR 84 2013-2023; Rivera-Hernández & Shervette, 2024; Rivera-Hernández & Shervette, 2024 |
| Wahoo | – | – | Non-comprehensive | – | – |
| White grunt | – | – | Non-comprehensive | – | – |

Table 4. Comprehensive list of available life history data for priority species in St. Croix.

| | | | | AVAILABLE DATA | |
|---|---|---|---|---|---|
| *SPECIES* | *SEDAR* | *SEDAR DATA RECOMMENDATION* | *STATUS OF AVAILABLE DATA* | *REPRODUCTION* | *AGE / GROWTH* |
| Blackfin snapper | – | – | Non-comprehensive | – | – |
| Blue tang | SEDAR30 | Size composition<br>Age composition<br>Life history | Non-comprehensive | – | – |
| Bluestriped grunt | – | – | Non-comprehensive | – | – |
| Caribbean spiny lobster | SEDAR8<br><br>SEDAR46<br><br>SEDAR57<br><br><br>SEDAR91 | Mortality of juveniles<br>Size composition<br><br>Length composition<br>Life history<br><br>Stock structure and abundance<br>Size composition<br>Life history<br><br>[IN PROGRESS] | Non-comprehensive | – | – |
| Coney grouper | – | – | Non-comprehensive | – | – |
| Doctorfish | – | – | Non-comprehensive | – | – |
| Dolphinfish | – | – | Non-comprehensive | – | – |
| French angelfish | – | – | Non-comprehensive | – | – |
| Gray angelfish | – | – | Non-comprehensive | – | – |
| Honeycomb cowfish | – | – | Non-comprehensive | – | – |
| King mackerel | – | – | Non-comprehensive | – | – |
| Little tunny | – | – | Non-comprehensive | – | – |
| Mutton snapper | SEDAR14 | Size composition<br>Age composition<br>Life history | Non-comprehensive | – | – |
| Princess parrotfish | – | – | Non-comprehensive | Jones et al., 2021 | Jones et al., 2021 |
| Queen conch | SEDAR14 | Size composition<br>Age composition<br>Habitat | Non-comprehensive | SK 2024-2026 collections for life history | SK 2024-2026 collections for life history |

| | | | | | |
|---|---|---|---|---|---|
| Queen parrotfish | – | – | Comprehensive | Life history data available from 2013-2023; Rivera-Hernández & Shervette, 2024; Rivera-Hernández & Shervette, 2024 | Life history data available from 2013-2023; Rivera-Hernández & Shervette, 2024; Rivera-Hernández & Shervette, 2024 |
| Queen triggerfish | SEDAR30<br><br>SEDAR80 | Size composition<br>Age composition<br>Life history<br><br>[IN PROGRESS] | Comprehensive | Rivera-Hernández & Shervette, 2024 covers 2013-2023 | Rivera-Hernández & Shervette, 2024 covers 2013-2023 |
| Red hind grouper | SEDAR35 | Size composition<br>Age composition<br>Life history | Non-comprehensive | – | – |
| Redband parrotfish | – | – | Comprehensive | Life history data available from 2013-2023; MARFIN 2021 report | Life history data available from 2013-2023; MARFIN 2021 report |
| Redfin parrotfish | – | – | Comprehensive | Life history data available from 2013-2023; MARFIN 2021 report | Life history data available from 2013-2023; MARFIN 2021 report |
| Redtail parrotfish | SEDAR26 | Size composition<br>Age composition<br>Age and growth<br>Life history | Comprehensive | Life history data available from 2013-2023; MARFIN 2021 report; currently processing 2022-2023 samples | Life history data available from 2013-2023; MARFIN 2021 report; currently processing 2022-2023 samples |
| Scrawled cowfish | – | – | Non-comprehensive | – | – |
| Stoplight parrotfish | SEDAR46<br><br>SEADR84 | Length composition<br>Life history<br><br>[IN PROGRESS] | Comprehensive | Life history data available from 2013-2023; SEDAR 84 2013-2023; Rivera-Hernandez & Shervette, 2024; Rivera- Hernandez & Shervette, 2024 | Life history data available from 2013-2023; SEDAR 84 2013-2023; Rivera-Hernandez & Shervette, 2024; Rivera-Hernandez & Shervette, 2024 |
| White grunt | – | – | Non-comprehensive | – | – |
| Yellowfin tuna | – | – | Non-comprehensive | – | – |
| Yellowtail snapper | SEDAR8 | Growth<br>Life history<br>Mortality<br>Size composition<br>Age composition | Non-comprehensive | – | – |

Table 5. Life history data needs for priority species in the U.S. Caribbean. X indicates no data are available. O indicates some data are available but more are still needed. — indicates the species is not a priority for that island platform. Blanks indicate data are available. Species for which data availability was listed as "comprehensive" in Tables 2–4 were removed from this table.

| *SPECIES* | *PUERTO RICO DATA NEEDS* | | *ST. THOMAS / ST. JOHN DATA NEEDS* | | *ST. CROIX DATA NEEDS* | |
|---|---|---|---|---|---|---|
| | *REPRODUCTION* | *AGE / GROWTH* | *REPRODUCTION* | *AGE / GROWTH* | *REPRODUCTION* | *AGE / GROWTH* |
| Bar jack | O | X | — | — | — | — |
| Blackfin snapper | — | — | X | X | X | X |
| Blackfin tuna | X | X | — | — | — | — |
| Blue runner | O | X | X | X | — | — |
| Blue tang | — | — | X | X | X | X |
| Bluestriped grunt | — | — | — | — | X | X |
| Boxfishes | O | O | X | X | — | — |
| Caribbean spiny lobster | X | X | X | X | X | X |
| Cero mackerel | O | X | X | X | — | — |
| Coney grouper | O | X | X | X | X | X |
| Doctorfish | — | — | X | X | X | X |
| Dolphinfish | O | X | X | X | X | X |
| French angelfish | — | — | — | — | X | X |
| Gray angelfish | — | — | X | X | X | X |
| Graysby grouper | O | X | X | X | — | — |
| Hogfish | | | X | X | — | — |
| Honeycomb cowfish | — | — | — | — | X | X |
| King mackerel | O | X | X | X | X | X |
| Lane snapper | | | | | — | — |
| Little tunny | X | X | X | X | X | X |
| Mutton snapper | O | X | X | X | X | X |
| Princess parrotfish | — | — | — | — | X | X |
| Queen angelfish | — | — | X | X | — | — |
| Queen conch | O | O | O | O | O | O |
| Queen parrotfish | | | — | — | | |
| Queen snapper | O | O | O | O | — | — |

| | | | | | | |
|---|---|---|---|---|---|---|
| Queen triggerfish | | | | | | |
| Rainbow parrotfish | X | X | — | — | — | — |
| Rainbow runner | X | X | X | X | — | — |
| Red hind grouper | O | X | O | O | X | X |
| Redband parrotfish | — | — | — | — | | |
| Redfin parrotfish | — | — | — | — | | |
| Redtail parrotfish | | | | | | |
| Scrawled cowfish | — | — | X | X | X | X |
| Silk snapper | O | O | — | — | — | — |
| Skipjack tuna | X | X | — | — | — | — |
| Stoplight parrotfish | | | | | | |
| Wahoo | O | X | X | X | — | — |
| White grunt | O | O | X | X | X | X |
| Yellowfin tuna | X | X | — | — | X | X |
| Yellowtail snapper | | | — | — | X | X |

## B. Availability of Life History Data:

1. Puerto Rico Life History Data Availability

   Among the priority species list for Puerto Rico, seven species (hogfish, lane snapper, queen parrotfish, queen triggerfish, redtail parrotfish, stoplight parrotfish, and yellowtail snapper) have comprehensive regionally-specific life history data available for use in stock assessment, 15 species have some or temporally limited data available, and seven do not have comprehensive data available. The following priority species have at least some reproductive data available: bar jack, blue runner, cero mackerel, coney grouper, graysby grouper, king mackerel, mutton snapper, red hind grouper, and wahoo. The following priority species/groups have at least some reproductive and age/growth data available: boxfishes, dolphinfish, hogfish, lane snapper, queen conch, queen parrotfish, queen snapper, queen triggerfish, redtail parrotfish, silk snapper, stoplight parrotfish, white grunt, and yellowtail snapper. The following priority species do not have life history data available: blackfin tuna, Caribbean spiny lobster, little tunny, rainbow parrotfish, rainbow runner, skipjack tuna, and yellowfin tuna.

2. St. Thomas/St. John Life History Data Availability

   Among the priority species list for St. Thomas/St. John, four species (lane snapper, queen triggerfish, redtail parrotfish, and stoplight parrotfish) have comprehensive regionally-specific life history data available for use in stock assessment, three species have some or temporally limited data available, and 24 do not have comprehensive data available. The following priority species have at least some reproductive and age/growth data available: lane snapper, queen conch, queen snapper, queen triggerfish, red hind grouper, redtail parrotfish, stoplight parrotfish. The following priority species/groups do not have life history data available: blackfin snapper, blue runner, blue tang, boxfishes, Caribbean spiny lobster, cero mackerel, coney grouper, doctorfish, dolphinfish, gray angelfish, graysby grouper, hogfish, king mackerel, little tunny, margate grunt, mutton snapper, queen angelfish, rainbow runner, scrawled cowfish, wahoo, and white grunt.

3. St. Croix Life History Data Availability

   Among the priority species list for St. Croix, six species (queen parrotfish, queen triggerfish, redband parrotfish, redfin parrotfish, redtail parrotfish, and stoplight parrotfish) have comprehensive regionally-specific life history data available for use in stock assessment, one species has some or temporally limited data available, and 19 do not have comprehensive data available. The following priority species have at least some reproductive and age/growth data available: queen conch, queen parrotfish, queen triggerfish, redband grouper, redfin parrotfish, redtail parrotfish, and stoplight parrotfish. The following priority species/groups do not have life history data available: blackfin tuna, blue tang, bluestriped grunt, Caribbean spiny lobster, coney grouper, doctorfish, dolphinfish, french angelfish, gray angelfish, honeycomb cowfish, king mackerel, little tunny, mutton snapper, princess parrotfish, red hind grouper, scrawled cowfish, white grunt, yellowfin tuna, and yellowtail snapper.

## C. Specific Recommendations for Life History Data Needs:

1. Puerto Rico Life History Data Needs

   Life history data from Puerto Rico for priority species are not currently comprehensive. In order to obtain informative management advice from stock assessment methods, additional biological data are essential. It is important that reproductive and age and growth data are collected island wide from Puerto Rico for the following priority species: bar jack, blackfin tuna, blue runner, boxfishes, Caribbean spiny lobster, cero mackerel, coney grouper, dolphinfish, graysby grouper, king mackerel, little tunny, mutton snapper, queen conch, queen snapper, rainbow parrotfish, rainbow runner, red hind grouper, silk snapper, skipjack tuna, wahoo, white grunt, and yellowfin tuna. For the species in which currently available life history data may be sufficient for a successful stock assessment, continued reproductive and age and growth data collections should still continue to inform future management actions. Those priority species include hogfish, lane snapper, queen parrotfish, queen triggerfish, redtail parrotfish, stoplight parrotfish, and yellowtail snapper.

2. St. Thomas / St. John Life History Data Needs

   Life history data from St. Thomas/St. John for priority species are not currently comprehensive. In order to obtain informative management advice from stock assessment methods, additional biological data are essential. It is important that reproductive and age and growth data are collected island wide from St. Thomas/St. John for the following priority species: blackfin snapper, blue runner, blue tang, boxfishes, Caribbean spiny lobster, cero mackerel, coney grouper, doctorfish, dolphinfish, gray angelfish, graysby grouper, hogfish, king mackerel, little tunny, margate grunt, mutton snapper, queen angelfish, queen conch, queen snapper, rainbow runner, red hind grouper, scrawled cowfish, wahoo, and white grunt. For the species in which currently available life history data may be sufficient for a successful stock assessment, continued reproductive and age and growth data collections should still continue to inform future management actions. Those priority species include lane snapper, queen triggerfish, redtail parrotfish, and stoplight parrotfish.

3. St. Croix Life History Data Needs

   Life history data from St. Croix for priority species are not currently comprehensive. In order to obtain informative management advice from stock assessment methods, additional biological data are essential. It is important that reproductive and age and growth data are collected island wide from St. Croix for the following priority species: blackfin snapper, blue tang, bluestriped grunt, Caribbean spiny lobster, coney grouper, doctorfish, dolphinfish, french angelfish, gray angelfish, honeycomb cowfish, king mackerel, little tunny, mutton snapper, princess parrotfish, queen conch, red hind grouper, scrawled cowfish, white grunt, yellowfin tuna, and yellowtail snapper. For the species in which currently available life history data may be sufficient for a successful stock assessment, continued reproductive and age and growth data collections should still continue to inform future management actions. Those priority species include queen parrotfish, queen triggerfish, redband parrotfish, redfin parrotfish, redtail parrotfish, and stoplight parrotfish.

## Appendix


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